Normalgon Modular Character Pack

Intro

Fully modular character system for Unity!

- Animation ready (tested with Mixamo, Humanoid)
- Swap head, hair, torso, legs, arms, socketed items, etc
- Body shape and colors are customizable, eg armor accents, skin, eyes, etc

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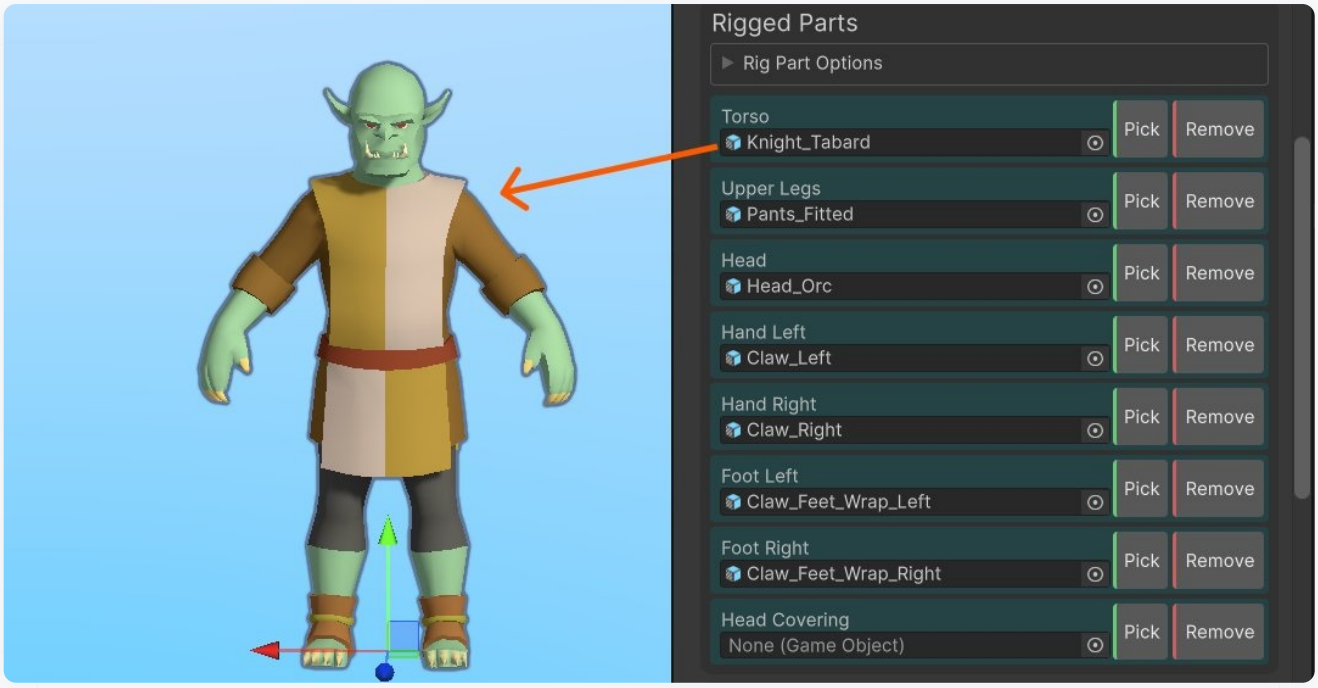
Questions, Comments, or Feature Requests?

Fully assembled modular character in action

Quickstart

1. Open the `Modular Character Demo Scene`
2. Select any character
3. In the inspector, view the `Modular Character` component
4. Use the UI here to swap parts, colors, and socketed items - you can even do this live, with the game running and animations working!
5. Duplicate and make prefabs for your game as you need



Character hierarchy in Unity showing bone structure and socketed items

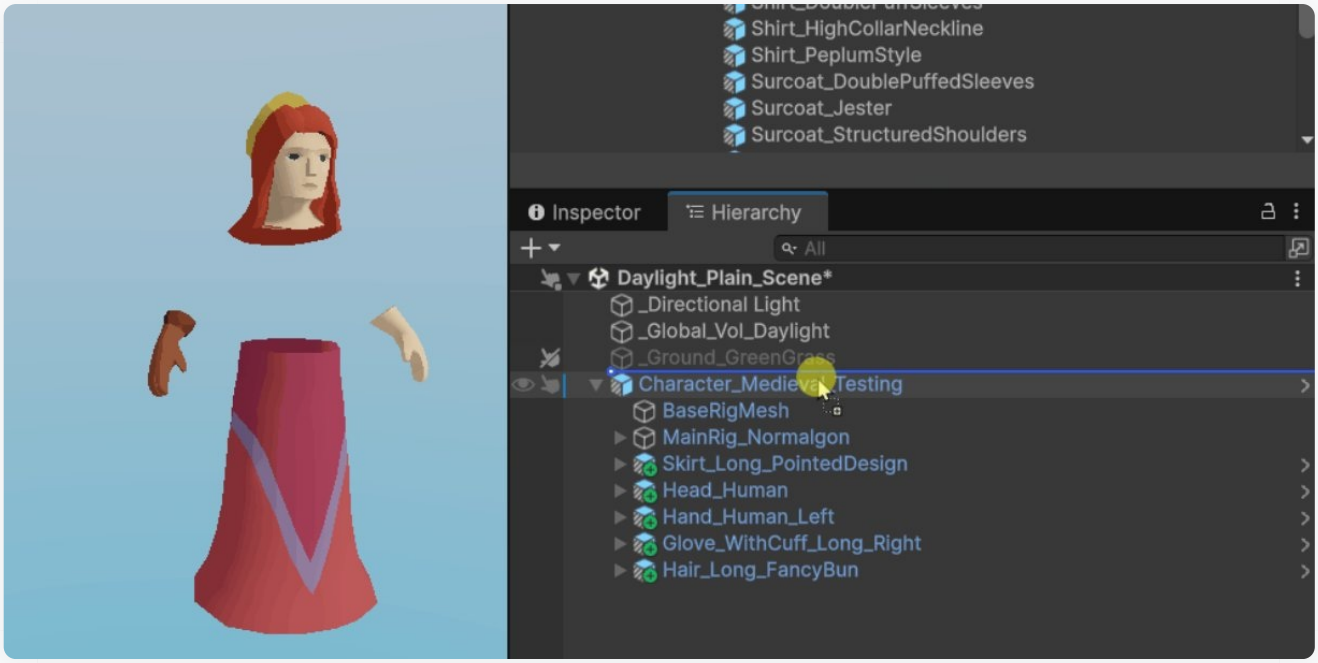
DIY: Basic Usage

> **Note:** If you prefer automated, see the **Modular Character Editor section**.

This section describes how to use the Character parts directly, manually.

Assembling and Testing a Character

1. Open the Project View
2. Navigate to `Normalgon / Modular Characters / Starter Set` (or the set of your choice)
3. Drag-drop the `Character_Starter` prefab to your scene
4. Drag-drop parts (eg, torso, head, left arm, etc) into the scene, as children of the `Character_Starter` prefab
    a. Note: Check each part - it should have a position of (0,0,0), but different versions of Unity can change this
5. Select the `Character_Starter`, and apply an Animation to test with
    a. No animations, or not sure how? We recommend **This Simple Tutorial**



Individual character parts before assembly

That's all! Hit play, and your assembled character will animate perfectly! The key here is the `Bone Remapper` script on the `Character_Starter`. It ensures all the separate parts animate together as one.

Customizing Colors

You can use any material on these characters, but we also provide a custom-built material for setting specific color zones on characters.

Rigged parts selection interface

1. Open the Project View
2. Navigate to `Normalgon / Shared Assets`
3. Duplicate the `M_Normalgon_CharacterColor` material into your own folder
4. Drag-drop the material onto your character parts to apply it
5. Select the material, and view the Inspector settings
6. As you can see, there are lots of options here! Try any color combos you like, and use the buttons to quick-set values
7. If you'd like an area of a character to be shiny, set its color alpha above .5, then the `Smoothness` and `Metallic` sliders will apply to that area

This all works via simple, performant vertex coloring and shaders. Have fun creating truly unique characters!

Body Shapes

All shapes, sizes, and combinations!

Color grid showing customizable zones for character materials

These body sections are modifiable via simple sliders:

- Head: `Face` to blend from square/masculine to tapered/feminine
- Torso: `Hips`, `Waist`, `Bust`, modify small to large
- Upper Legs: `Hips`, `Waist`, modify small to large
- Feet: `Feet`, modify small to large

We expect to add more of these over time.

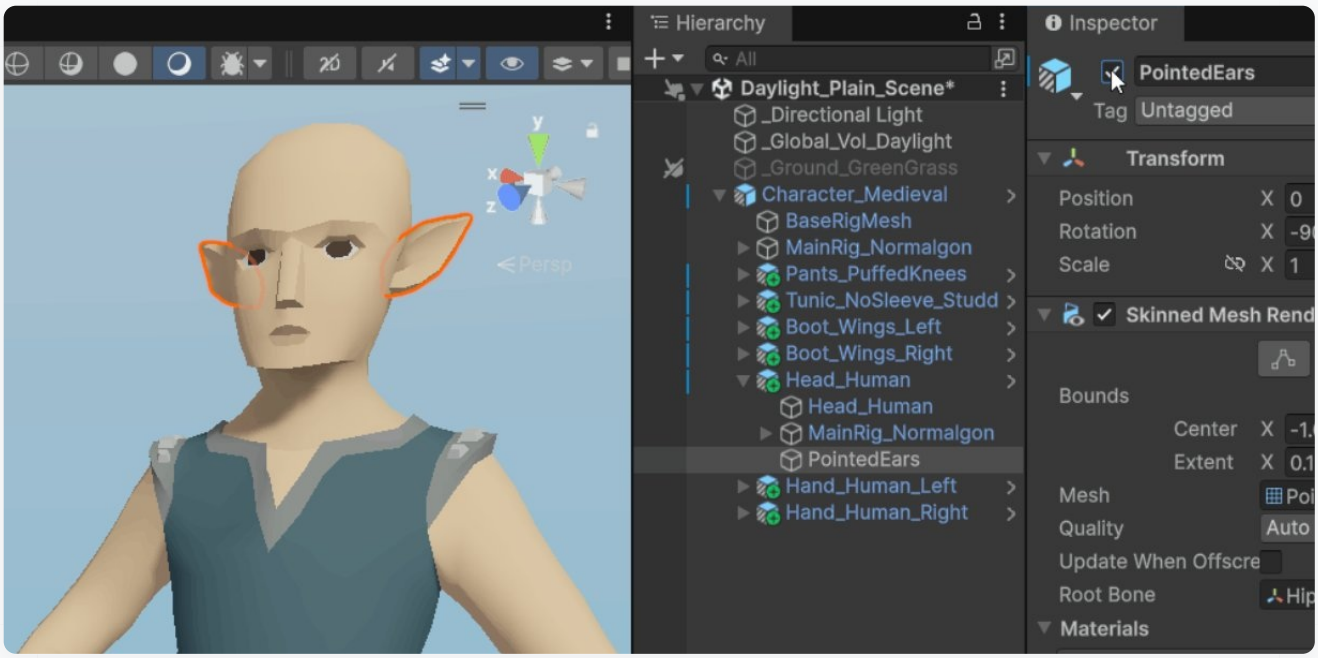
> **Note:** The `Waist` value should be the same on both Torso and Upper Legs, this is where the models attach.

To edit a shape:

1. Select the object to edit
2. Find the `Skinned Mesh Renderer` component (may be in a child object)
3. Use the slider(s) under `BlenderShapes`

Pointed Ears Toggle

Human and fantasy character heads include the ability to toggle pointed ears on or off directly in the prefab. This is useful for creating elves, fantasy races, or other characters with distinctive ear shapes.



The PointedEars child object can be enabled or disabled in the head prefab hierarchy

To toggle pointed ears:

1. Select the character's head in the hierarchy (e.g., `Head_Human`)
2. Expand the head object to view its children
3. Find the `PointedEars` child object
4. Enable or disable the `PointedEars` object using the checkbox in the Inspector

Sockets And Accessories

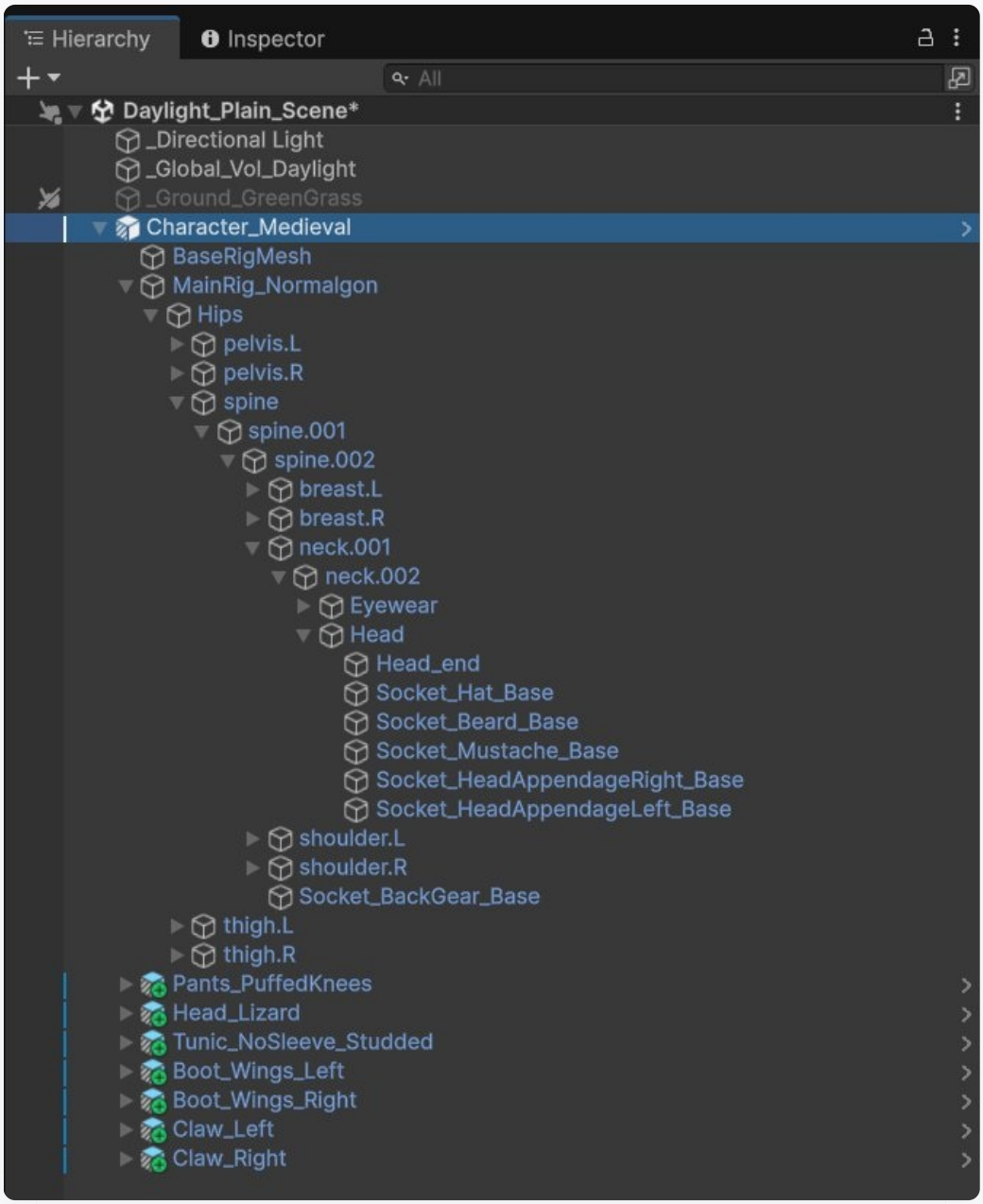
Armor, backpacks, hats, even hats on hats - you can have it all.

> **Note:** we highly recommend using the automated Modular Character Editor, since manual socket setup is tedious and difficult.

To manually add a socketed item:

1. Open the Project View

2. Navigate to `Normalgon / Modular Characters / Starter Set` (or the set of your choice)
3. Drag-drop any Socketed Item to be children of a socket in the character you are building



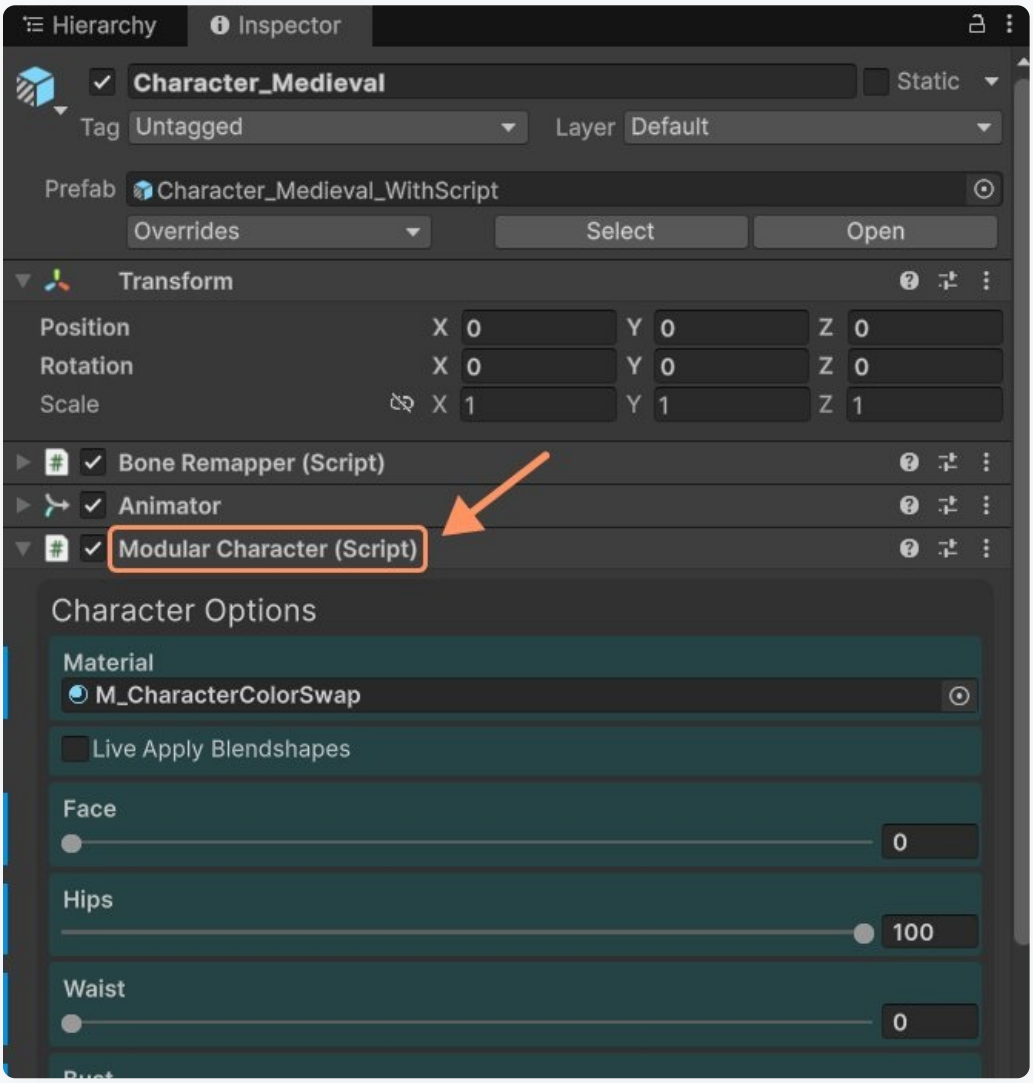
BlendShapes interface showing Hips, Waist, and Bust sliders for body customization

Automated: Modular Character Editor

The Modular Character Editor was designed to make character editing simple, fast, and game-ready. We use it for all our demos and tests, it works both in-editor and at runtime (eg, in your game).

Setup

1. Open the Project View
2. Navigate to `Normalgon / Modular Characters / Starter Set` (or the set of your choice)
3. Drag-drop the `Character_Medieval` (or other) into the scene
4. In the Inspector, scroll to the `Modular Character` component

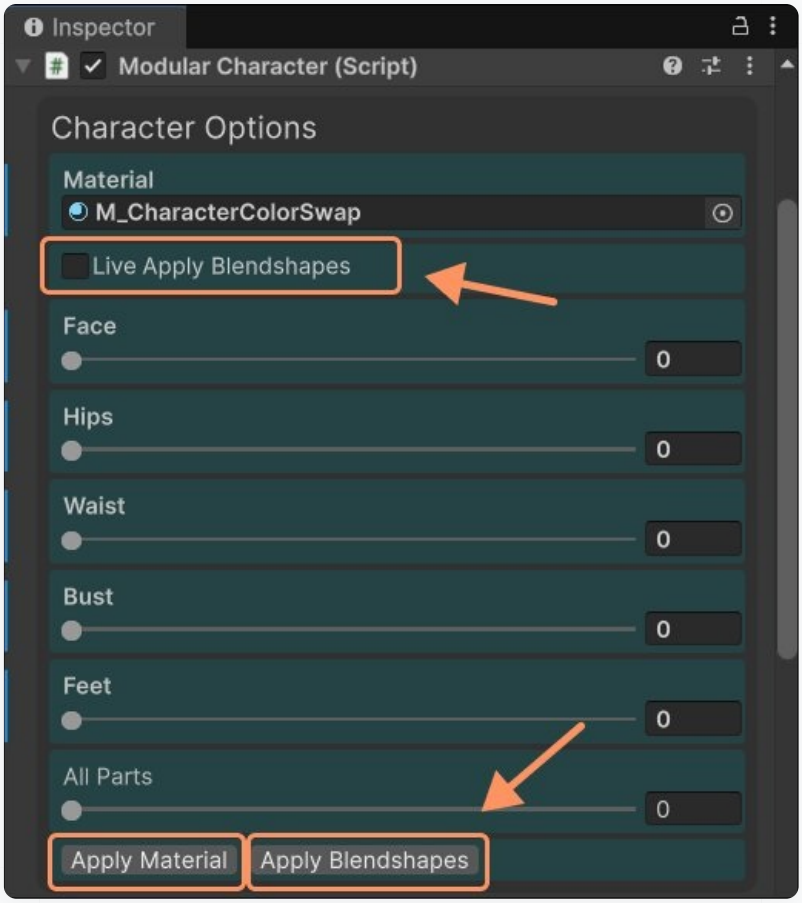


The Modular Character Script component in the Unity Inspector

Character Options Section

Properties and settings that affect the entire character

- `Material` : the material to use across the character parts. Use the bottom "Apply Material" button to apply this to all parts.
- `Live Apply Blendshapes` : if ON, modifying a blendshape will update live on all character parts. Otherwise, use the "Apply Blendshapes" button.

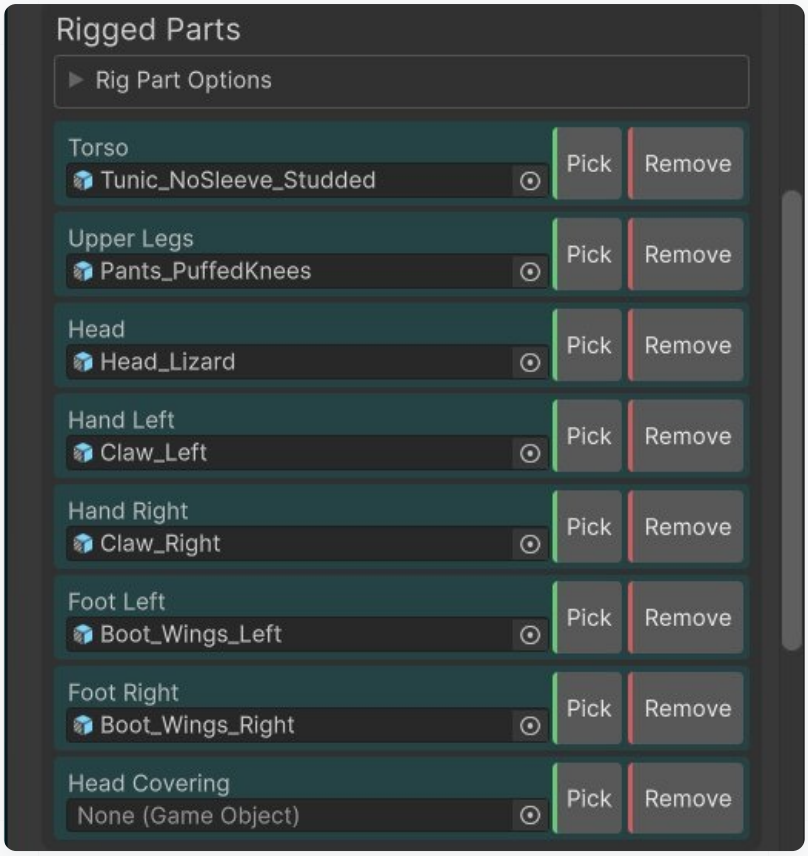


Character Options showing Material settings and blendshape controls

Rigged Parts

Parts that are rigged, eg move/twist/deform with animation.

- `Rigged Part Options` : modify the parts available to each section below
- `Torso` : Shows the current torso part, and "Pick", "Remove" buttons for quickly setting or removing the part, using the Rig Part Options noted above
- (same for `Upper legs` , `Head` , `Hand Left` , `Hand Right` , `Foot Left` , `Foot Right` , `Head Covering`)

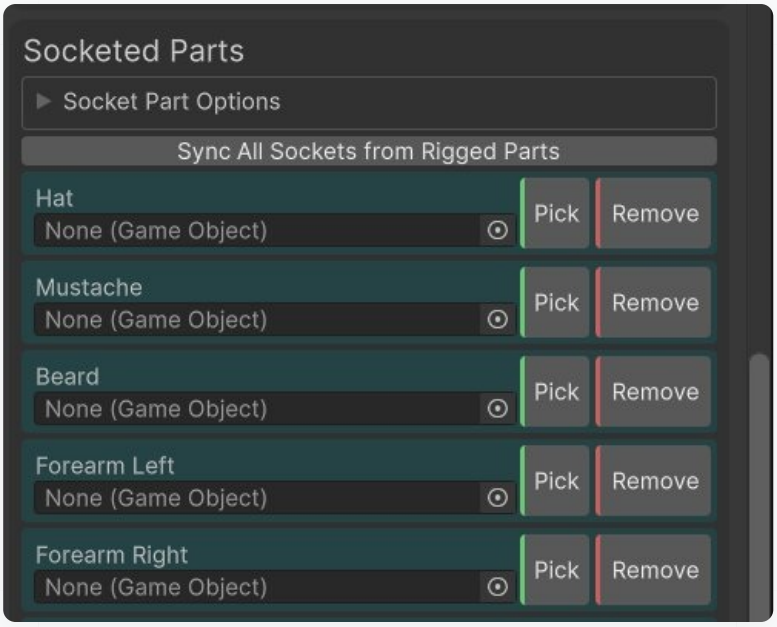


Rigged Parts interface showing all body part options with Pick and Remove buttons

Socketed Parts

Parts that do not deform, but should move with the character

- `Socket Part Options` : modify the parts available to each section below
- `Sync All Sockets from Rigged Parts` - use this if you manually change rigged parts, to ensure sockets are at the correct locations
- `Hat` , `Mustache` , `Beard` , etc - all of these will spawn in and place the item exactly where it needs to be, with exact offset and parenting. Use the Pick and Remove buttons.



Socketed Parts interface for accessories like hats, mustaches, beards, and forearms

Troubleshooting Mode

If something isn't working as you expect here, enable the toggle here and start a chat with us! We'll be happy to help, and this will show us logs to know exactly what is happening.

Enable Debug Logs for detailed console output when diagnosing issues

Questions, Comments, or Feature Requests?

We're always building more packs and options, and love to see your creations too. Visit the website for social, discord, and other links.

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